Cell Passaging

C01 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Maintain HEK293T, U251, SH-SY5Y and CCF-STTG1 cultures by trypsinization and reseeding. The medium recipes, usual digestion times, split ratios and culture intervals below describe our routine maintenance workflow; adjust the timing of passage to the observed culture condition.

Materials & Equipment

Item	Supplier / specification	Use
DMEM; RPMI 1640	Gibco; established laboratory formulations	Choose the basal medium listed below
Fetal bovine serum (FBS)	Cellmax	50 mL per 500 mL basal medium
Penicillin-Streptomycin (PS)	Gibco; 100× stock	5 mL per 500 mL basal medium
Trypsin-EDTA	Gibco; 0.25%, 1×	Ready-to-use cell dissociation reagent [2]
PBS	Gibco; sterile, without Ca2+ or Mg2+	Wash before trypsinization
Culture vessels; tubes; pipettes and tips	Axygen tips; laboratory supplies	Collection, resuspension and reseeding
Cell culture cabinet; centrifuge; microscope	Eppendorf centrifuge; laboratory equipment	Aseptic handling, pelleting and inspection
Humidified CO2 incubator	37°C, 5% CO2	Routine culture of the listed cells

Before You Start

Prepare the two complete media

Mix the ingredients aseptically in a sterile container that holds approximately 555 mL. Label medium type, preparation date and supplement lots. Store according to the component instructions; prewarm the working aliquot.

Complete medium	Basal medium	Cellmax FBS	Gibco PS (100×)	Total
HEK293T / U251 / SH-SY5Y	500 mL DMEM	50 mL	5 mL	555 mL
CCF-STTG1	500 mL RPMI 1640	50 mL	5 mL	555 mL

The 500:50:5 recipe gives approximately 9.01% FBS and 0.901% PS stock (0.901×) in the final mixture. These are additions to 500 mL basal medium, not final 10% FBS plus 1% PS. PS stock strength is specified in [3].

RPMI 1640 is the basal-medium name for CCF-STTG1. Our Gibco/Cellmax recipe above is the laboratory formulation; ATCC's reference formulation specifies ATCC-formulated RPMI 1640 with final 10% FBS. [4]

Prepare for passage

Inspect confluence, morphology and medium appearance before starting. Passage actively growing cultures before crowding compromises their condition. Prewarm the required complete medium and prepare PBS and the working trypsin aliquot; label receiving vessels with cell line, passage number and date. [1]

Procedure

A. Wash and detach cells

1. Aspirate the spent medium. Gently rinse the monolayer once with Ca2+/Mg2+-free PBS, using enough to cover the surface, then remove the wash completely. Direct liquid along the vessel wall to avoid stripping loosely attached cells. [1]

2. Add Gibco 0.25% Trypsin-EDTA (1×) to cover the cell layer in a thin, continuous film. As an area-based reference, approximately 0.5 mL per 10 cm^2 can be used; confirm complete coverage. Start timing when the enzyme contacts the monolayer. [1]

3. Incubate at 37°C using the usual digestion times below as observation checkpoints. Inspect for cell rounding and release; neutralize once cells detach instead of extending digestion merely to complete the nominal time.

Cell line	Usual time at 37°C	Routine split	Usual interval / decision
HEK293T (293T)	1 min 30 s	1:4	Approximately every 48 h
U251	3 min	1:4	Approximately every 72 h
SH-SY5Y	2 min 30 s	1:3	Variable condition; passage according to morphology and confluence
CCF-STTG1	5 min	1:3	Usually needs passage after approximately 72 h in our cultures

4. Add at least twice the trypsin volume of the matching serum-containing complete medium to stop digestion. Gently pipette to collect released cells and disperse loose clumps, then transfer to a sterile centrifuge tube. [1]

B. Collect and reseed

5. Centrifuge at 200 × g for 5 min at room temperature as a general working condition. Carefully remove the supernatant without disturbing the pellet, then gently resuspend in a measured volume of fresh complete medium. [1]

6. Mix the suspension evenly and transfer the fraction indicated by the split ratio into each fresh vessel of the same growth area. A 1:4 split means one quarter of the recovered cells per new vessel; 1:3 means one third. Add fresh complete medium to the normal working volume.

Calculation examples: resuspend all cells in 4 mL and take 1 mL for a 1:4 split, or in 3 mL and take 1 mL for 1:3. For a different growth area or quantitative experiment, count cells and set the required density.

Culture vessel	Approximate medium volume
6-well plate	1-3 mL per well
T25 flask	3-5 mL
T75 flask	8-15 mL

Vessel volumes are reference ranges; select within the vessel manufacturer's guidance and keep the chosen volume consistent between routine passages. [5]

7. Distribute cells with gentle back-and-forth and side-to-side motion. Incubate at 37°C with 5% CO2, using appropriate gas exchange for the vessel. Check attachment and morphology the next day.

8. Monitor cultures daily and schedule the next passage using the table as a planning guide. Record the actual digestion time, split ratio and passage interval. If cells reach the appropriate density earlier, passage earlier; if they remain sparse or unhealthy, assess their condition before splitting again.

p.s.

Have neutralizing medium ready before adding trypsin, especially for the 1 min 30 s 293T digestion. Process only as many vessels as you can inspect and stop promptly.

For SH-SY5Y, retain the usual 1:3 split but judge the passage day from morphology and confluence; its condition has been variable in our cultures.

Our CCF-STTG1 experience differs from the cell-bank guidance we received: at 1:3, cultures usually need passage again after approximately 72 h. Monitor daily. This local observation is not a measured doubling time or a universal growth characteristic.

Keep medium formulation, serum lot and trypsin conditions consistent. PS does not replace aseptic technique or contamination checks.

Expected Outcome

Cells should reattach, show the characteristic morphology of the culture and resume growth. The actual time to the next passage should broadly follow the table when starting condition and vessel area are comparable, but no fixed growth rate or viability threshold is implied. Retain cell-line identity, passage number, medium batch and observations of unusual morphology or detachment.

Troubleshooting

Observed problem	Check first	Suggested action
Cells remain attached at the usual time	Residual serum/PBS, enzyme coverage, temperature and activity	Inspect microscopically and continue only as needed. If detachment remains poor, check reagent handling rather than using forceful pipetting. [1, 2]
Excess debris or poor reattachment	Digestion duration, delay before neutralization and mechanical handling	Stop promptly at detachment in later passages; neutralize immediately and reduce harsh pipetting. Review culture health before repeating the split.
Dense clumps or uneven distribution	Completeness of detachment and mixing before aliquoting	Disperse loose clumps gently, remix before each aliquot and distribute cells evenly after plating. Avoid trying to break resistant clumps by vigorous trituration.
Culture becomes crowded earlier than expected	Split arithmetic, vessel growth area and starting density	Passage according to observed density. For CCF-STTG1 at 1:3, approximately 72 h is our usual experience; do not wait for a longer assumed schedule.
SH-SY5Y remains sparse or changes morphology	Starting health, detachment damage, medium/serum consistency and contamination	Reassess before another 1:3 split; allow recovery where appropriate and investigate persistent abnormalities.
Cloudy medium or suspicious particles	Microscopy, aseptic technique and shared reagents	Separate the culture and investigate under the laboratory contamination procedure; do not simply increase PS.

Reference

[1] Thermo Fisher Scientific / Gibco. Adherent Cell Culture Protocol: washing, enzyme coverage, neutralization, pelleting and reseeding. Accessed 2026-10-03. Open source

[2] Gibco. Trypsin-EDTA (0.25%), 1×, product-family page 25200056 and handling FAQs. Accessed 2026-10-03. Open source

[3] Gibco. Penicillin-Streptomycin, 100×: 10,000 U/mL penicillin and 10,000 µg/mL streptomycin; product page 15140122. Accessed 2026-10-03. Open source

[4] ATCC. CCF-STTG1, CRL-1718: RPMI-1640 reference medium and culture conditions. Accessed 2026-10-03. Open source

[5] Thermo Fisher Scientific / Gibco. Useful Numbers for Cell Culture: medium-volume ranges for 6-well plates, T25 and T75 flasks. Accessed 2026-10-03. Open source